

Sabastian Asuo-Darko

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Professional Summary

Research-focused Biomedical Engineer with First-Class Honours and a peer-reviewed publication in pressure ulcer prevention, underpinning future smart medical device development. Applied computational modelling (ANSYS, MATLAB, SolidWorks) to study pulmonary mechanics in ventilator-induced lung injury for undergraduate thesis, translating physiological insight into simulation-driven design. Combines clinical engineering experience with computational biomechanics expertise to advance medical device innovation from research to real-world impact.

Curious and coachable professional

Research Interests

- Computational Biomechanics & Physiological Modelling
- Medical Device Design, Simulation & Validation
- Respiratory Mechanics & Ventilator-Induced Lung Injury (VILI)
- Biomedical Systems Analysis & Translational Engineering

Publications & Projects

Causal Factors and Their Effects on the Human Body for the Design of a Smart Bedsores Prevention System

Sept 2023 – Jan 2024

Co-authored Publication — IJITIS

KNUST, Ghana

- Co-authored a multidisciplinary study analysing pressure, temperature, and humidity as key contributors to pressure ulcer formation.
- Supported system design, data analysis, and modelling to inform the development of a smart bedsores prevention device.
- Research addresses a clinical challenge affecting 2.5M+ patients annually, with projected reduction potential of up to 40% in high-risk groups.
- [View Publication](#) [🔗](#)

Simulating Pulmonary Mechanics

Jan 2024 – Aug 2024

Undergraduate Thesis — Computational Modelling of VILI

KNUST, Ghana

- Led a two-person research project modelling spatial pressure propagation in lung tissue to study ventilator-induced lung injury (VILI).
- Designed anatomically inspired 3D lung models and ran ANSYS simulations to evaluate tissue response under varying ventilation pressures.
- Analysed simulation outputs to identify stress distribution trends relevant to ventilator control optimisation.
- Findings contribute to improved mechanical ventilation strategies for ICU patients requiring respiratory support.
- [View Documentation](#) [🔗](#)

Design and Finite Element Analysis of an Above-Knee Prosthetic Leg Biomedical Engineering Design Project

Jan 2022

KNUST, Ghana

- Designed a low-cost above-knee prosthetic leg using the biomedical engineering design process.
- Conducted market analysis, functional decomposition, and QFD to define engineering specifications.
- Modelled prosthetic components and full assembly in SolidWorks.
- Performed FEA to assess stress, displacement, and strain under physiological loading.
- Evaluated material selection and proposed validation tests for safety and durability.
- [View Documentation](#) [🔗](#)

Computational Design and Testing of Sports Equipment Engineering Simulation Project

June 2024

KNUST, Ghana

- Applied computational modelling techniques to analyse stress distribution and performance characteristics.
- Used simulation-driven insights to guide iterative design improvements.
- [View Documentation](#) [🔗](#)

Industry and Clinical Experience

Machine Medics

Mar 2026 – Present

Field Service & Sales Engineer

Accra, Ghana

- Installed and commissioned 4+ Vacure Lyofia fluorescent immunoassay analysers and 2+ ECG machines across diagnostic centres, ensuring operational readiness and regulatory compliance.
- Performed preventive and corrective maintenance on diagnostic systems including Zybion haematology analysers, maintaining consistent device uptime.
- Delivered hands-on clinical training for laboratory and clinical staff, achieving user proficiency and reducing post-installation support calls.
- Lead technical presentations, proposals, and product comparisons during tenders and procurement processes.
- Manage customer accounts, maintain relationships with key stakeholders, and support contract renewals and equipment upgrades.

Sysmex West and Central Africa
Field Service Engineer

Oct 2024 – Feb 2026

Cantonments, Ghana

- Aided pre-installation site readiness for 30+ haematology and urinalysis analysers, cutting setup delays by 30%.
- Assisted in installation, configuration, and commissioning of diagnostic systems, achieving full operational readiness within one day per site.
- Participated in monthly preventive maintenance for 20+ analysers, contributing to a 20% reduction in equipment downtime.
- Provided weekly technical support for 3–5 devices, maintaining >95% uptime across assigned sites.
- Documented service activities and supported compliance with manufacturer and safety standards.

Sept 2023 – Dec 2023 & Sept 2021 – Dec 2021

University of Ghana Medical Centre & Ministry of Health —
Biomedical Engineering Unit
Biomedical Engineering Intern

Legon & Ridge, Ghana

- Performed routine maintenance on 20+ medical devices monthly at University of Ghana Medical Centre, reducing failure rates by approximately 25%.
- Repaired 10+ critical devices (suction units, infusion pumps), preventing operating room and ward delays.
- Helped servicing and safe usage of 5+ surgical systems, including laser machines and electrosurgical units.
- Tested and validated 30+ medical devices across 10 hospitals at Ministry of Health, ensuring 100% compliance with national standards.
- Logged device usage and performance data, improving asset tracking efficiency by 25% and supported national asset verification across 5+ regions.

More info on industry experience here: [Professional Portfolio](#)

Education

Jan 2021 – Nov 2024

Kwame Nkrumah University of Science & Technology (KNUST)
BSc. Biomedical Engineering — First Class Honours

Kumasi, Ghana

CWA 77.75/100 · Top 10% of cohort · Winner, BMESS Day of Code (July 2023) · Deputy Judicial and Electoral Committee Head, BMESS (2022–2024)

Software Competencies

Engineering & Simulation
ANSYS, MATLAB, LabVIEW

Productivity & Reporting
Microsoft Office Suite, Google Workspace, ZohoOne

Programming & Data
Python, C, C++, SQL, AI & ML

Other Tools
Git, ImageJ, Power BI, Adobe Suite

Design & Analysis
SolidWorks, AutoCAD, Blender

Certifications

Medical Device Essentials — SternMed GmbH · WHO Covid-19 Respiratory Equipment · Sysmex Caresphere User Operation Training · ALX AI Career Essential Course · Professional Development: IT Security & AI

Languages: English (Native) · French (Conversational)